

PlasmaXAI: Counterfactual-Guided Explainable Fusion for Plasma Cell Recognition in Multiple Myeloma

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Abstract. Recognizing malignant plasma cells is important for multiple myeloma screening, yet manual microscopic review remains slow. This paper presents PlasmaXAI, an explainable digital pathology framework that combines frozen ResNet50 and DenseNet121 embeddings with handcrafted morphology, a morphology-derived counterfactual boundary model, and learned fusion gates that inject counterfactual information directly into the final decision pathway. Evaluated on PCMMMD, PlasmaXAI achieved 93.42% accuracy, 93.43% weighted F1, 91.24% plasma precision, 94.63% plasma recall, and 97.76% AUC at the selected operating point. The framework also produced counterfactual, calibration, ablation, and exploratory patient-level analyses. These findings position PlasmaXAI as an interpretable screening framework for digital hematopathology.

Keywords: Multiple myeloma · Digital pathology · Explainable AI · Counterfactual analysis · Plasma cell recognition · Deep learning

1 Introduction

Multiple myeloma is a hematologic malignancy characterized by abnormal plasma-cell proliferation in bone marrow, and morphologic assessment remains central to diagnosis, disease monitoring, and treatment planning [1]. In many laboratories, stained marrow smears are still reviewed manually. This workflow is labor-intensive, sensitive to stain variation, and vulnerable to reader-to-reader disagreement.

Recent progress in computational pathology suggests that deep learning can reduce this burden, but strong accuracy alone is not sufficient for clinical use [2,3,12]. A clinically useful plasma-cell system should support review rather than replace it; therefore, it must expose why it predicts malignancy and which features would change the decision.

PlasmaXAI was developed with that requirement in mind. Instead of attaching explanations after prediction, the framework integrates counterfactual morphology information into a learned fusion network so that explainability contributes directly to classification. The study makes three main contributions. First, it introduces an explainable multi-branch architecture for malignant

plasma-cell recognition that combines deep image embeddings and interpretable morphology. Second, it embeds counterfactual distance and feature-shift signals inside the fusion pathway rather than treating them as post-hoc artifacts. Third, it evaluates the system beyond conventional accuracy using ablation, calibration, bootstrap confidence intervals, and patient-level interpretation analysis.

2 Literature Review

Deep learning has become the dominant approach in digital pathology because convolutional models can capture cell texture, chromatic variation, and structural context more effectively than handcrafted descriptors alone [2,11]. Studies in computational oncology show that clinically useful pathology systems must balance fine local detail with robust higher-level semantics across varying staining conditions [3,12]. Residual networks support deeper optimization through identity mappings, while densely connected networks improve feature reuse and gradient propagation [4,5]. These backbone families are therefore well suited to plasma-cell images, where subtle nuclear irregularity, chromatin behavior, and cytoplasmic staining all contribute to malignancy recognition.

Explainability research in medical AI has progressed through local surrogates, saliency maps, attribution methods, and counterfactual reasoning [13,14,15,6]. Counterfactual explanations are particularly attractive in healthcare because they answer a clinically intuitive question: what minimal change would move a prediction toward a different class? DICE extends this idea by generating diverse counterfactual sets that improve interpretive coverage [7]. Calibration is another important requirement, because a probability score has clinical value only when confidence aligns with observed frequency [8]. Few systems make counterfactual information part of the decision path itself. PlasmaXAI addresses that gap by combining deep image features, interpretable morphology, and counterfactual-guided learned fusion in one screening framework.

3 Materials and Methods

3.1 Dataset, Preprocessing, and Morphology Features

Experiments were conducted on PCMMMD using a cell-level classification set and a separate patient-organized cohort for exploratory patient analysis. Cell images were resized to 224×224 for CNN feature extraction. In parallel, a handcrafted morphology pipeline computed nucleus-to-cytoplasm ratio, nucleus area, cytoplasm area, roundness, staining intensity, granularity, and mean RGB channels. These descriptors form an interpretable phenotype layer that complements the learned image embeddings and supports counterfactual analysis.

3.2 PlasmaXAI Architecture

PlasmaXAI uses frozen ResNet50 and DenseNet121 encoders as complementary image branches. A logistic morphology model estimates a decision boundary

in handcrafted feature space, from which the framework derives counterfactual distance and feature-shift terms that summarize how far a sample lies from the benign region and which morphology changes would most strongly alter the decision.

These signals are passed to a learned fusion module that separately encodes ResNet, DenseNet, morphology, counterfactual, and score streams. Counterfactual features modulate the image branches through sigmoid gates and also contribute to a softmax-based modality gate that controls the final weighted fusion. This design makes counterfactual reasoning structurally relevant to prediction rather than merely descriptive after prediction. Figure 1 presents the overall workflow and Figure 2 shows the internal architecture in larger detail.

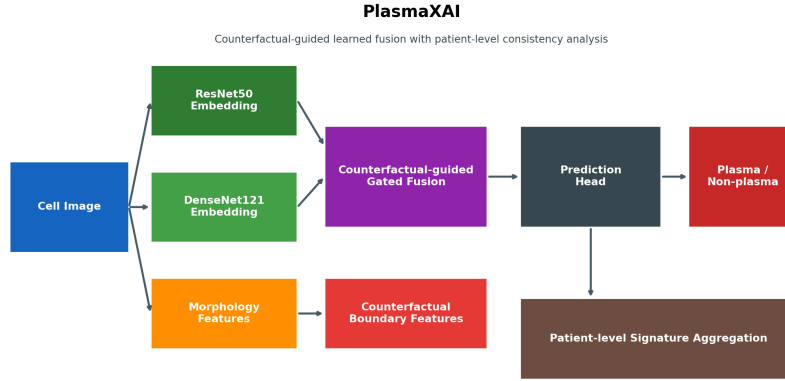


Fig. 1. End-to-end PlasmaXAI workflow from image preprocessing to explainable prediction.

Figure 1 outlines the full processing pipeline used to generate predictions, explanations, and patient-level outputs.

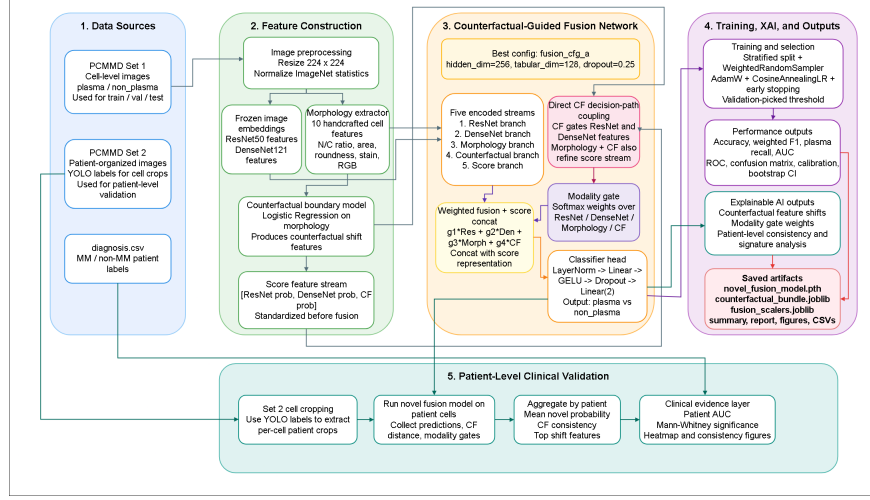


Fig. 2. Detailed internal PlasmaXAI fusion architecture with image, morphology, and counterfactual branches.

Figure 2 emphasizes that counterfactual features are inserted into the fusion process itself rather than appended only after classification.

3.3 Training and Evaluation

The deployment operating point was selected on validation data using a precision-recall balanced criterion. Final evaluation reports accuracy, weighted F1, plasma precision, plasma recall, and AUC. The study also includes ablation, bootstrap confidence intervals, calibration curves, and patient-level interpretation analysis.

4 Results and Discussion

4.1 Benchmark Performance

At the selected operating point, PlasmaXAI achieved 93.42% accuracy, 93.43% weighted F1, 91.24% plasma precision, 94.63% plasma recall, and 97.76% AUC. It therefore surpassed the strongest non-PlasmaXAI baselines on the three most deployment-relevant classification criteria: accuracy, precision, and recall. ResNet50 remained the strongest single deep backbone, while the previous optimized hybrid remained the strongest non-PlasmaXAI precision baseline.

Table 1. Held-out test performance for the strongest comparison models.

Model	Acc.	F1	Prec.	Rec.	AUC
ResNet50	91.92	91.93	89.02	93.80	97.05
Prev Hybrid	91.73	91.73	90.91	90.91	97.12
DenseNet121	89.85	89.83	90.17	87.19	96.36
PlasmaXAI	93.42	93.43	91.24	94.63	97.76

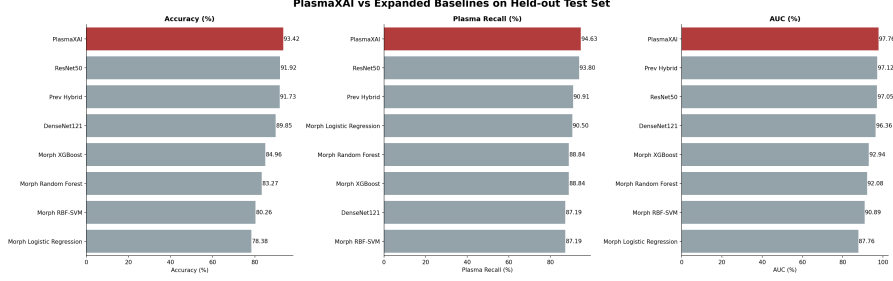


Fig. 3. Expanded benchmark comparison showing PlasmaXAI against deep and morphology-based baselines.

Figure 3 shows that the full framework delivers the strongest balance across the main benchmark metrics rather than improving one measure at the expense of another.

Bootstrap analysis indicated that the selected operating point was stable, with a 95% confidence interval of 91.16–95.30 for accuracy, 91.30–97.20 for plasma recall, and 96.56–98.63 for AUC. This matters because a reliable screening aid should not depend on one favorable split.

4.2 Explainability and Ablation

The explainability pathway is a defining characteristic of PlasmaXAI. Counterfactual analysis identifies the morphology shifts most responsible for malignant predictions while also showing how the learned gates redistribute attention across image and tabular streams. In practical terms, the system produces both a malignant probability and an explanation of which feature changes would move the case toward a benign boundary.

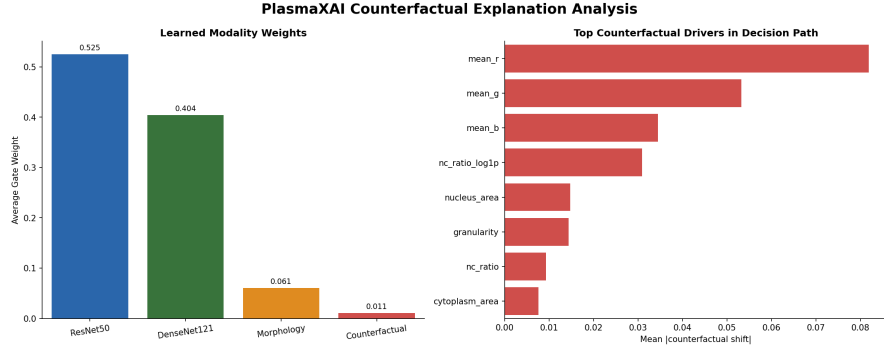


Fig. 4. Counterfactual explanation analysis for PlasmaXAI.

Figure 4 shows the feature-shift directions and modality-gating behavior that support a malignant decision.

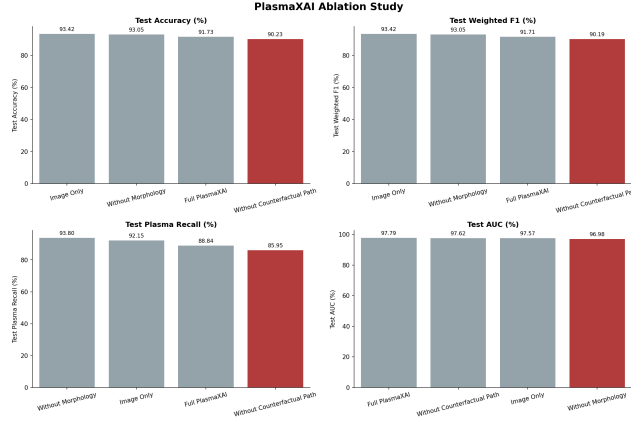


Fig. 5. Ablation study showing the effect of removing key PlasmaXAI components.

Figure 5 shows that removing the counterfactual branch causes the largest drop in malignant-cell recall.

The ablation results support this interpretation. Removing the counterfactual path reduced malignant recall to 85.95%, far below the full model, while removing morphology also degraded performance. These results suggest that the primary gain comes from structured interaction between deep image features and counterfactual morphology reasoning, not from any single CNN backbone alone.

4.3 Reliability and Patient-Level Interpretation

Internal reliability analysis remained favorable, and the patient-level patterns remained visually consistent across the exploratory cohort. Even so, the patient-level result should be interpreted carefully because the cohort is small and internally sourced. It is therefore presented as supportive evidence rather than definitive clinical validation.

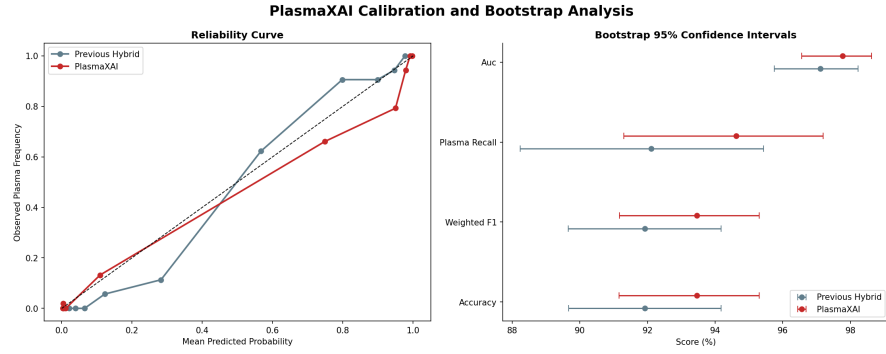


Fig. 6. Calibration curve and bootstrap stability analysis for PlasmaXAI.

Figure 6 shows that the selected operating point remains well behaved under repeated resampling.

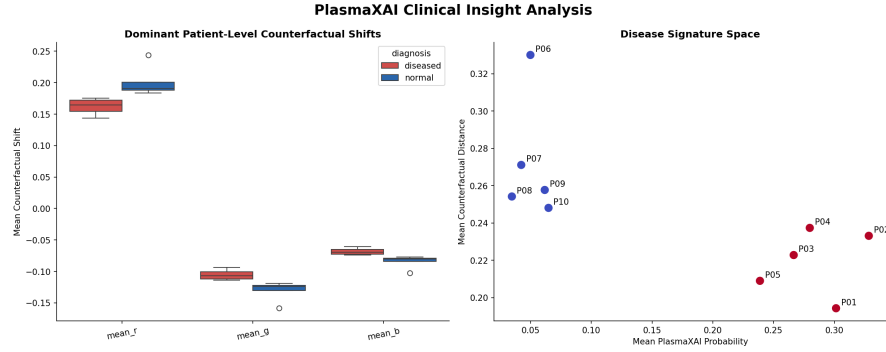


Fig. 7. Clinical insight diagram summarizing the dominant disease-signature patterns discovered by PlasmaXAI.

Figure 7 summarizes the morphologic and chromatic patterns most associated with the malignant decision path.

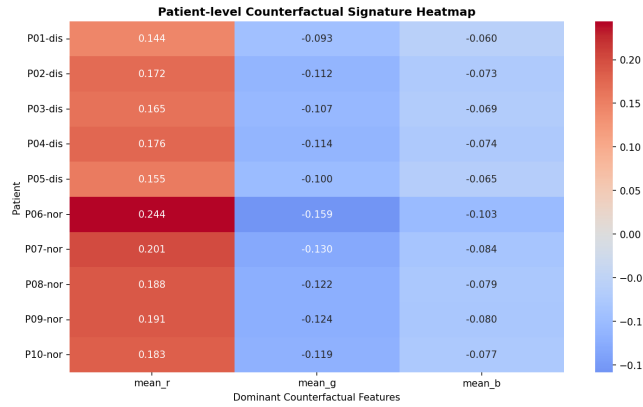


Fig. 8. Patient-level heatmap showing the consistency of disease-related morphology patterns across cases.

Figure 8 makes the patient-wise pattern consistency easier to inspect than a single scalar performance score.

Taken together, the results position PlasmaXAI as an explainable screening system rather than a narrowly optimized classifier. Its value lies in combining strong discrimination with traceable counterfactual reasoning and patient-wise consistency analysis.

5 Conclusion

This study presents PlasmaXAI as an explainable fusion framework for malignant plasma-cell recognition in multiple myeloma. By integrating deep image embeddings, interpretable morphology descriptors, and counterfactual signals inside a learned fusion pathway, the model achieves strong held-out performance while preserving transparency. PlasmaXAI outperforms key baselines, shows clear dependence on its counterfactual branch during ablation, and provides calibration, explanation, and patient-level analyses that support clinician-in-the-loop review. The main limitation is the lack of external multi-center validation. Future work should therefore emphasize stain-robust generalization, larger patient cohorts, and prospective hematopathology evaluation.

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